MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

Third Semester B.Sc(Hons) Degree Examination
MATHEMATICS(DSE)
MG3DSEMAT201- Game Theory and Project Management

[2024 Admissions Onwards]

Time: 2 Hours Marks 70

PART-A
Answer any 5 questions. Each question carries 2 Marks.

1. What do you mean by an optimal strategy in a game? [K][1]
2. Explain the term saddle point [K][1]
3. Find the value of the game

	Player B	
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$
$\mathbf{A_1}$	1	-1/2
$\mathbf{A_2}$	-1/2	0

[A][2]

4. Write different types of methods to find a solution for games with out saddle points? [U][2]
5. Discuss merge event and burst event with examples [K][3]
6. Write a short note on AOA and AON network diagrams. [U][3]
7. Write the difference between critical path and a critical activity. [U][4]
8. What do you understand when the total float value is negative? [U][4]

PART-B
Answer any 5 questions. Each question carries 6 Marks.

9. Examine the following payoff matrix for saddle points. $\begin{bmatrix} 0 & 2 & -3 & 0 \\ -2 & 0 & 0 & 0 \\ 3 & 0 & 0 & -4 \\ 0 & -3 & 4 & 0 \end{bmatrix}$ [U][1]
10. Explain the Rules of Dominance with proper examples [U][1]
11. Solve the following game

	Player B		
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$
$\mathbf{A_1}$	1	7	2
$\mathbf{A_2}$	6	2	7
$\mathbf{A_3}$	5	1	6

[A][2]

12. Find the value of the following game

	Player B	
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$
$\mathbf{A_1}$	3	7
$\mathbf{A_2}$	1	2
$\mathbf{A_3}$	5	-3

[A][2]

13. Explain different types of floats on activities [K][3]

14. Discuss different phases in a network. [K][3]

15. Explain Forward Pass Method. [U][4]

16. An architect has been awarded a contract to prepare a plans for an urban renewal project.The job consists of the following activities and estimated times.

Activity	Description	Duration(Weeks)	Predecessors
A	Prepare preliminary sketches	2	-
B	Outline specifications	1	-
C	Prepare drawings	3	A
D	Write specifications	2	A,B
E	Run off prints	1	C,D
F	Have specifications	3	B,D
G	Assemble bid packages	1	E,F

Find the critical path. [A][4]

PART-C
Answer any 3 questions . Each question carries 10 Marks.

17. Consider the game with following pay off table

	Player B	
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$
$\mathbf{A_1}$	2	6
$\mathbf{A_2}$	-2	λ

(a) Show that the game is strictly determinable, whatever λ may be

(b) Determine the value of the game

[U][1]

18. For the following payoff matrix, transform the zero-sum game into an equivalent linear programming problem and solve it by simplex method.

	Player B		
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$
$\mathbf{A_1}$	1	-1	3
$\mathbf{A_2}$	3	5	-3
$\mathbf{A_3}$	6	2	-2

[A][2]

19. Find the value of the game an optimal strategies of the the following game with graphical method

	Player B				
Player A	$\mathbf{B_1}$	$\mathbf{B_2}$	$\mathbf{B_3}$	$\mathbf{B_4}$	$\mathbf{B_5}$
$\mathbf{A_1}$	3	7	5	-1	3
$\mathbf{A_2}$	1	2	3	-1	-9

[An][2]

20. A new type of water pump is to be designed for an automobile.Major specifications are given in the table. Draw the network diagram of activities involved in the project.

Activity	Description	Predecessor Activity
A	Drawing prepared and approved	-
B	Cost analysis	A
C	Tool feasibility	A
D	Tool manufactured	C
E	Favourite cost	B,C
F	Raw materials produced	D,E
G	Sub assemblies ordered	E
H	Sub assemblies received	G
I	Parts manufactured	D,F
J	Final assembly	I,H
K	Testing and shipment	I

[U][3]

21. Following table gives the list of various activities and their immediate predecessors involved in the CAT scanner in a hospital.

Activity	Description	Predecessors	Duration(days)
A	Finalization of the layout plans	-	2
B	Demolition of the structure	A	6
C	Walls erection	B	12
D	Flooring	B	8
E	Electrical wiring	C	6
F	Air conditioning ducting	C	4

G	Fire alarm installation	C	3
H	False ceiling and light fittings	E,F,G	10
I	Wall plastering and painting	H	9
J	Equipment installation	D,I	6
K	Calibration and testing	J	3
L	Final finishing	K	2
M	Handing over	L	1

(a) Prepare a network diagram showing the interrelationship of the various activities.

(b) Find the total time required to hold the conference.

(c) Compute the total float for the noncritical activities.

[An][4]

22. Listed in the table are the activities and their simplified sequencing requirements for publishing a textbook.

Activity	Description	Predecessors	Expected completion time(Months)
A	Write book	-	12
B	Design book	A	1
C	Edit manuscript	A	6
D	checking editing	C	2
E	Accept design	B	1
F	Copy edit	D,E	2
G	Prepare artwork	D,E	4
H	Accept and correct artwork	G	1
I	Set galleys	F	4
J	Check and correct galleys	I	1
K	Full page proofs	H,J	2
L	Check and correct pages	K	1
M	Prepare index	K	1
N	Set and correct index	M	1
O	Check camera-ready copy	L,N	1
P	Print and bind books	O	1

(a) Construct the network of activities and find the critical path.

(b) For each non-critical activity,find independent float and free float.

[An][4]